

Similar to p-IgGen latents, IgLM and ProGen2-OAS TopK latents correlated to IGHJ4 activate at or near the J-region of antibodies (Supplementary Figure 6). None of the top IgLM or ProGen2-OAS IGHJ features demonstrate a strong correlation to aligned positions as their corresponding p-IgGen features. This may be explained by low feature consistency with TopK SAEs (Wang et al., 2024), where the nature of features identified is sensitive to the architecture of the base language model.

Finally, to assess whether steering interventions transfer across model architectures, we identified Ordered SAE features strongly correlated with specific IGHJ identities. For both IgLM and ProGen2-OAS, we positively steered on a low-index feature associated with IGHJ6 (Figure 3).

For ProGen2-OAS steering, correlations between the target IGHJ6 score and the steered latent activation were highly significant (Pearson $r = 0.988$, $p = 6.661 \times 10^{-7}$; Spearman $\rho = 0.983$, $p = 1.936 \times 10^{-6}$). For IgLM steering, we likewise observed significant correlations, though with smaller magnitude (Pearson $r = 0.813$, $p = 4.038 \times 10^{-4}$; Spearman $\rho = 0.802$, $p = 5.563 \times 10^{-4}$).

Replicating key experiments on IgLM and ProGen2-OAS suggests that two qualitative findings are robust: (i) SAE latents can recover antibody-relevant gene-identity structure from internal activations, and (ii) Ordered SAEs reliably expose steerable features. These results support the view that SAE-based interpretability is not confined to a single architecture or scale, but can serve as a general-purpose diagnostic tool across diverse antibody language models.

5. Conclusions and Future Outlook

We show that SAEs can successfully be used to evaluate autoregressive antibody language models and identify learned domain-specific features. Current SAE implementations on pLMs for identifying features are limited by the assumption that feature correlation is equal to biological concept causation. Analysis of activated latents reveal high predictive performance does not always correspond to steerability.

We found that some latents correlated to gene identity simply mark conserved positions without a clear causal link to gene identity. Some highly correlated features may highlight non-specific aspects of a concept, but be correlated to more concrete features. When we attempted to steer p-IgGen by amplifying these individual latents, the resulting antibody libraries did not show a reliable increase in IGHJ4 usage, further underscoring that high predictive power alone doesn't guarantee steerability.

Ordered SAEs solve this by identifying hierarchical features which correspond to more abstract and higher-level concepts rather than simple residue-level identity. However, this comes at the cost of intuitive interpretability of activation patterns.

One of the major limitations of employing SAEs to antibody language models is the lack of labelled datasets, unlike general proteins (Suzek et al., 2007). This prevents automated feature identification and annotation, which has allowed for their easy application for general protein language models (PLMs) (Simon & Zou, 2024). Lack of functionally annotated data remains a problem within the

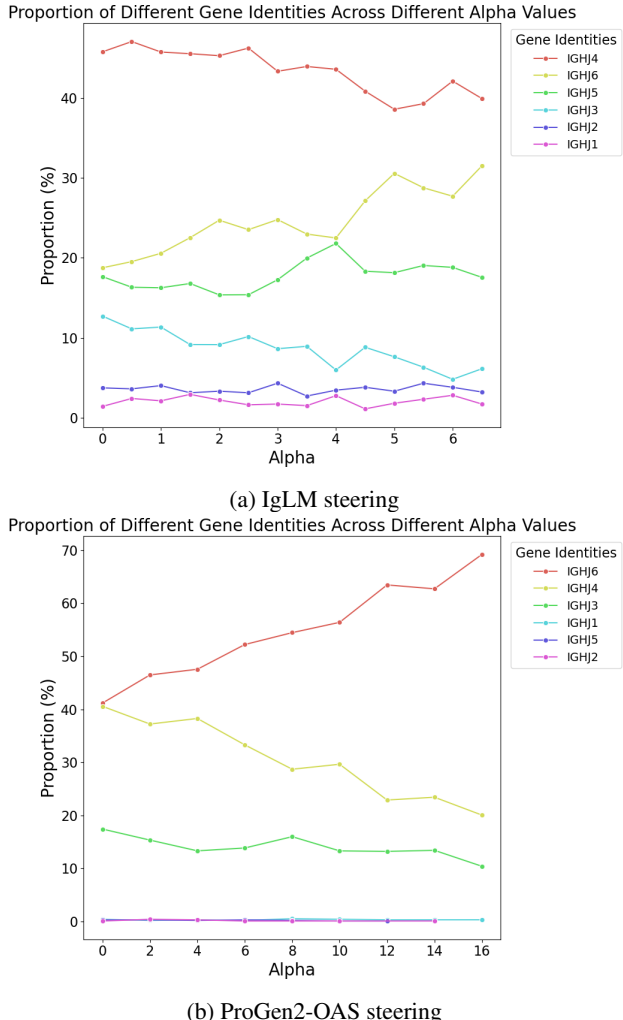


Figure 3. Results of IGHJ feature steering for IgLM latent, positively correlated to IGHJ4 (a), and ProGen2-OAS latent, positively correlated to IGHJ6 (b). The y-axis shows the proportion of generated sequences. Plots are coloured by heavy J gene identity. The x-axis shows the steering factor used (α). Results are for a library of 1000 generated sequences.

domain of antibody language models.

Several issues remain to enable the use of SAEs for PLM interpretability. For example, exhaustive steering analyses should be conducted to quantitatively measure the steerability of Top-K and Ordered SAE features and SAE training should be scaled using more expansive, annotated datasets like FLAb (Chungyoun et al., 2024) to enable models to identify domain-specific abstractions and facilitate targeted manipulation of generative outputs. This would enable the generation of antibody libraries with properties such as developability and specificity, improving modern library generation methods and optimising the antibody drug development pipeline.

References

- Abanades, B., Olsen, T. H., Raybould, M. I. J., Aguilar-Sanjuan, B., Wong, W. K., Georges, G., Bujotzek, A., and Deane, C. M. The Patent and Literature Antibody Database (PLAbDab): an evolving reference set of functionally diverse, literature-annotated antibody sequences and structures. *Nucleic Acids Research*, 52(D1):D545–D551, January 2024. ISSN 0305-1048. doi: 10.1093/nar/gkad1056. URL <https://dx.doi.org/10.1093/nar/gkad1056>. Publisher: Oxford Academic.
- Adams, E., Bai, L., Lee, M., Yu, Y., and AlQuraishi, M. From Mechanistic Interpretability to Mechanistic Biology: Training, Evaluating, and Interpreting Sparse Autoencoders on Protein Language Models, February 2025. URL <https://www.biorxiv.org/content/10.1101/2025.02.06.636901v1>. Pages: 2025.02.06.636901 Section: New Results.
- Andreano, E. and Rappuoli, R. Immunodominant antibody germelines in COVID-19. *The Journal of Experimental Medicine*, 218(5):e20210281, March 2021. ISSN 0022-1007. doi: 10.1084/jem.20210281. URL <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7933983/>.
- Bussmann, B., Nabeshima, N., Karvonen, A., and Nanda, N. Learning Multi-Level Features with Matryoshka Sparse Autoencoders, March 2025. URL <http://arxiv.org/abs/2503.17547>. arXiv:2503.17547 [cs].
- Capel, H. L., Ellmen, I., Murray, C. J., Mignone, G., Black, M., Clarke, B., Breen, C., Tierney, S., Dougan, P., Buick, R. J., Greenshields-Watson, A., and Deane, C. M. LICHEN enables light-chain immunoglobulin sequence generation conditioned on the heavy chain and experimental needs. *Communications Biology*, February 2026. ISSN 2399-3642. doi: 10.1038/s42003-026-09727-3. URL <https://www.nature.com/articles/s42003-026-09727-3>.
- Chanin, D., Wilken-Smith, J., Dulka, T., Bhatnagar, H., and Bloom, J. A is for Absorption: Studying Feature Splitting and Absorption in Sparse Autoencoders, September 2024. URL <http://arxiv.org/abs/2409.14507>. arXiv:2409.14507 [cs].
- Chen, J.-Y., Wang, J.-F., Hu, Y., Li, X.-H., Qian, Y.-R., and Song, C.-L. Evaluating the advancements in protein language models for encoding strategies in protein function prediction: a comprehensive review. *Frontiers in Bioengineering and Biotechnology*, 13, January 2025. ISSN 2296-4185. doi: 10.3389/fbioe.2025.1506508. URL <https://www.frontiersin.orghttps://www.frontiersin.org/journals/bioengineering-and-biotechnology/articles/10.3389/fbioe.2025.1506508/full>. Publisher: Frontiers.
- Chen, W., Liu, X., Zhang, S., and Chen, S. Artificial intelligence for drug discovery: Resources, methods, and applications. *Molecular Therapy Nucleic Acids*, 31:691–702, March 2023. ISSN 2162-2531. doi: 10.1016/j.omtn.2023.02.019. URL [https://www.cell.com/molecular-therapy-family/nucleic-acids/abstract/S2162-2531\(23\)00039-2](https://www.cell.com/molecular-therapy-family/nucleic-acids/abstract/S2162-2531(23)00039-2). Publisher: Elsevier.
- Chinery, L., Jeliakov, J. R., and Deane, C. M. Humatch - fast, gene-specific joint humanisation of antibody heavy and light chains, September 2024. URL <https://www.biorxiv.org/content/10.1101/2024.09.16.613210v1>. Pages: 2024.09.16.613210 Section: New Results.
- Chiu, M. L., Goulet, D. R., Teplyakov, A., and Gilliland, G. L. Antibody Structure and Function: The Basis for Engineering Therapeutics. *Antibodies*, 8(4):55, December 2019. ISSN 2073-4468. doi: 10.3390/antib8040055. URL <https://www.mdpi.com/2073-4468/8/4/55>. Number: 4 Publisher: Multidisciplinary Digital Publishing Institute.
- Chungyoun, M., Ruffolo, J., and Gray, J. FLAb: Benchmarking deep learning methods for antibody fitness prediction, January 2024. URL <https://www.biorxiv.org/content/10.1101/2024.01.13.575504v1>. Pages: 2024.01.13.575504 Section: New Results.
- Crescioli, S., , Hélène, K., , Lin, W., , Jyothsna, V., , Vaishali, K., , and Reichert, J. M. Antibodies to watch in 2025. *mAbs*, 17(1):2443538, December 2025. ISSN 1942-0862. doi: 10.1080/19420862.2024.2443538. URL <https://doi.org/10.1080/19420862.2024.2443538>. Publisher: Taylor & Francis .eprint: <https://doi.org/10.1080/19420862.2024.2443538>.
- Deng, W., Niu, X., He, P., Yan, Q., Liang, H., Wang, Y., Ning, L., Lin, Z., Zhang, Y., Zhao, X., Feng, L., Qu, L., and Chen, L. An allelic atlas of immunoglobulin heavy chain variable regions reveals antibody binding epitope preference resilient to SARS-CoV-2 mutation escape. *Frontiers in Immunology*, 15:1471396, January 2025. ISSN 1664-3224. doi: 10.3389/fimmu.2024.1471396. URL <https://www.frontiersin.org/journals/immunology/articles/10.3389/fimmu.2024.1471396/full>. Publisher: Frontiers.
- Dunbar, J. and Deane, C. M. ANARCI: antigen receptor numbering and receptor classification. *Bioinformatics*, 32(2):298–300, January 2016. ISSN 1367-4803. doi: 10.1093/bioinformatics/btv552. URL <https://doi.org/10.1093/bioinformatics/btv552>.
- Gao, L., Tour, T. D. I., Tillman, H., Goh, G., Troll, R., Radford, A., Sutskever, I., Leike, J., and Wu, J. Scaling and evaluating sparse autoencoders, June 2024. URL <http://arxiv.org/abs/2406.04093>. arXiv:2406.04093 [cs] version: 1.
- Jarasch, A., Koll, H., Regula, J. T., Bader, M., Papadimitriou, A., and Kettenberger, H. Developability Assessment During the Selection of Novel Therapeutic Antibodies. *Journal of Pharmaceutical Sciences*, 104(6):1885–1898, June 2015. ISSN 0022-3549, 1520-6017. doi: 10.1002/jps.24430. URL [https://jpharmsci.org/article/S0022-3549\(15\)30084-8/abstract](https://jpharmsci.org/article/S0022-3549(15)30084-8/abstract). Publisher: Elsevier.

- Kovaltsuk, A., Leem, J., Kelm, S., Snowden, J., Deane, C. M., and Krawczyk, K. Observed Antibody Space: A Resource for Data Mining Next-Generation Sequencing of Antibody Repertoires. *The Journal of Immunology*, 201(8):2502–2509, October 2018. ISSN 0022-1767. doi: 10.4049/jimmunol.1800708. URL <https://doi.org/10.4049/jimmunol.1800708>. eprint: <https://academic.oup.com/jimmunol/article-pdf/201/8/2502/61431152/ji1800708.pdf>.
- Kästner, L. and Crook, B. Explaining AI through mechanistic interpretability. *European Journal for Philosophy of Science*, 14(4):52, October 2024. ISSN 1879-4920. doi: 10.1007/s13194-024-00614-4. URL <https://doi.org/10.1007/s13194-024-00614-4>.
- Lefranc, M.-P., Pommié, C., Ruiz, M., Giudicelli, V., Foulquier, E., Truong, L., Thouvenin-Contet, V., and Lefranc, G. IMGT unique numbering for immunoglobulin and T cell receptor variable domains and Ig superfamily V-like domains. *Developmental and Comparative Immunology*, 27(1):55–77, January 2003. ISSN 0145-305X. doi: 10.1016/s0145-305x(02)00039-3.
- Li, W., Jaroszewski, L., and Godzik, A. Clustering of highly homologous sequences to reduce the size of large protein databases. *Bioinformatics (Oxford, England)*, 17(3):282–283, March 2001. ISSN 1367-4803. doi: 10.1093/bioinformatics/17.3.282.
- Marks, C., Hummer, A. M., Chin, M., and Deane, C. M. Humanization of antibodies using a machine learning approach on large-scale repertoire data. *Bioinformatics*, 37(22):4041–4047, November 2021. ISSN 1367-4803. doi: 10.1093/bioinformatics/btab434. URL <https://doi.org/10.1093/bioinformatics/btab434>.
- Nijkamp, E., Ruffolo, J. A., Weinstein, E. N., Naik, N., and Madani, A. ProGen2: Exploring the boundaries of protein language models. *Cell Systems*, 14(11):968–978.e3, November 2023. ISSN 2405-4712, 2405-4720. doi: 10.1016/j.cels.2023.10.002. URL [https://www.cell.com/cell-systems/abstract/S2405-4712\(23\)00272-7](https://www.cell.com/cell-systems/abstract/S2405-4712(23)00272-7).
- Olah, C., Cammarata, N., Schubert, L., Goh, G., Petrov, M., and Carter, S. Zoom In: An Introduction to Circuits. *Distill*, 5(3):10.23915/distill.00024.001, March 2020a. ISSN 2476-0757. doi: 10.23915/distill.00024.001. URL <https://distill.pub/2020/circuits/zoom-in>.
- Olah, C., Cammarata, N., Schubert, L., Goh, G., Petrov, M., and Carter, S. Zoom In: An Introduction to Circuits. *Distill*, 2020b. doi: 10.23915/distill.00024.001.
- Olsen, T. H., Boyles, F., and Deane, C. M. Observed Antibody Space: A diverse database of cleaned, annotated, and translated unpaired and paired antibody sequences. *Protein Science*, 31(1):141–146, 2022. ISSN 1469-896X. doi: 10.1002/pro.4205. URL <https://onlinelibrary.wiley.com/doi/abs/10.1002/pro.4205>. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/pro.4205>.
- Parsan, N., Yang, D. J., and Yang, J. J. Towards Interpretable Protein Structure Prediction with Sparse Autoencoders, March 2025a. URL <http://arxiv.org/abs/2503.08764>. arXiv:2503.08764 [q-bio].
- Parsan, N., Yang, D. J., and Yang, J. J. Towards Interpretable Protein Structure Prediction with Sparse Autoencoders. In *ICLR 2025 Workshop on Generative and Experimental Perspectives for Biomolecular Design (GEM)*, 2025b. URL <https://arxiv.org/abs/2503.08764>. eprint: 2503.08764.
- Prihoda, D., Maamary, J., Waight, A., Juan, V., Fayadat-Dilman, L., Svozil, D., and Bitton, D. A. BioPhi: A platform for antibody design, humanization, and humanness evaluation based on natural antibody repertoires and deep learning. *mAbs*, 14(1):2020203, December 2022. ISSN 1942-0862. doi: 10.1080/19420862.2021.2020203. URL <https://doi.org/10.1080/19420862.2021.2020203>. eprint: <https://doi.org/10.1080/19420862.2021.2020203>.
- Rai, D., Zhou, Y., Feng, S., Saparov, A., and Yao, Z. A Practical Review of Mechanistic Interpretability for Transformer-Based Language Models, March 2025. URL <http://arxiv.org/abs/2407.02646>. arXiv:2407.02646 [cs].
- Raybould, M. I. J. and Deane, C. M. The Therapeutic Antibody Profiler Therapeutic antibody profiler (TAP) for Computational Developability Assessment. In Houen, G. (ed.), *Therapeutic Antibodies: Methods and Protocols*, pp. 115–125. Springer US, New York, NY, 2022. ISBN 978-1-0716-1450-1. doi: 10.1007/978-1-0716-1450-1_5. URL https://doi.org/10.1007/978-1-0716-1450-1_5.
- Raybould, M. I. J., Kovaltsuk, A., Marks, C., and Deane, C. M. CoV-AbDab: the coronavirus antibody database. *Bioinformatics*, 37(5):734–735, May 2021. ISSN 1367-4803, 1367-4811. doi: 10.1093/bioinformatics/btaa739. URL <https://academic.oup.com/bioinformatics/article/37/5/734/5893556>.
- Scaviner, D., Barbié, V., Ruiz, M., and Lefranc, M.-P. Protein Displays of the Human Immunoglobulin Heavy, Kappa and Lambda Variable and Joining Regions. *Experimental and Clinical Immunogenetics*, 16(4):234–240, November 1999. ISSN 0254-9670. doi: 10.1159/000019115. URL <https://doi.org/10.1159/000019115>.
- Shuai, R. W., Ruffolo, J. A., and Gray, J. J. IgLM: Infilling language modeling for antibody sequence design. *Cell Systems*, 14(11):979–989.e4, November 2023. ISSN 2405-4720. doi: 10.1016/j.cels.2023.10.001.
- Simon, E. and Zou, J. InterPLM: Discovering Interpretable Features in Protein Language Models via Sparse Autoencoders, November 2024. URL <http://arxiv.org/abs/2412.12101>. arXiv:2412.12101 [q-bio].
- Suzek, B. E., Huang, H., McGarvey, P., Mazumder, R., and Wu, C. H. UniRef: comprehensive and non-redundant UniProt reference clusters. *Bioinformatics*, 23(10):1282–1288, May 2007. ISSN 1367-4803. doi: 10.1093/bioinformatics/btm098. URL <https://doi.org/10.1093/bioinformatics/btm098>.